

HITHESH K

Embedded Systems / Firmware Engineer

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate (2026) focused on embedded firmware and hardware-software integration. Built two end-to-end ESP32/Arduino systems combining sensor interfacing, real-time decision logic, UART, GPS/GSM and cloud connectivity, plus a real-time EV battery telemetry pipeline during an internship. Strong in Embedded C/C++, interrupts, timers, ADC/PWM, RTOS, networking (TCP/IP, MQTT) and VLSI design tools. Seeking Embedded, Firmware, BSW, VLSI or IoT roles.

TECHNICAL SKILLS

Core Proficiency: Embedded C/C++, ESP32 and Arduino firmware, sensor interfacing, UART serial communication, GPS/GSM integration, Cadence, VLSI design tools, AutoCAD

Languages & MCUs: Embedded C, C, C++, Python, Verilog | STM32, ESP32, ESP32-CAM, Arduino

Peripherals & Drivers: GPIO, timers, interrupts (ISR design), ADC, PWM, watchdog timer, DMA, memory-mapped I/O, register-level configuration

Firmware Architecture: startup/boot flow, memory layout (stack, heap, static/global, .data/.bss), RTOS (task scheduling, semaphores, mutexes, queues), state machines, bit manipulation, volatile/const usage, firmware debugging

Serial & Bus Protocols: UART, SPI, I2C, CAN, GPS and GSM module interfacing, sensor interfacing (analog and digital)

Networking & IoT: OSI and TCP/IP model, TCP vs UDP, IP addressing, sockets, HTTP/HTTPS, REST, WebSockets, MQTT (publish/subscribe, QoS), Wi-Fi, client-server and cloud data flow

EDA & Design Tools: Cadence, VLSI design software (RTL, simulation, synthesis flow), AutoCAD, PCB prototyping

Digital, Hardware & Tools: digital electronics, FSM design, RTL design, FPGA, hardware debugging, Git/GitHub, Linux, OpenCV, VS Code

PROJECTS

AI-Controlled Agricultural Rover

ESP32, ESP32-CAM, Embedded C/C++, Python, OpenCV, IoT, Cloud

- Designed an autonomous agriculture rover with a layered architecture: sensing, embedded control, camera-based perception, IoT connectivity and cloud monitoring.
- Programmed ESP32 firmware to read field sensors, execute rover control logic and send observations to cloud services over wireless links for remote monitoring.
- Integrated ESP32-CAM modules with a Python/OpenCV pipeline for crop disease detection, and built soil-analysis functions for field-level decisions, automating rover operations from sensor feedback.

Drunk Driver Detection & Vehicle Safety System

Arduino/ESP32, C/C++, MQ-3 sensor, GPS/GSM, OpenCV

- Built a multi-sensor vehicle-safety prototype that reads an MQ-3 alcohol sensor and runs OpenCV-based driver detection to identify impaired-driving conditions before the vehicle starts.
- Wrote embedded decision logic that fuses both inputs and drives an automated ignition-lock control, integrating detection, control and notification modules in real time.
- Interfaced GPS and GSM modules over serial (UART) to send location-aware emergency alerts.

INTERNSHIP EXPERIENCE

Hardware Engineering Intern – Smart EV Charging & Battery Twin Dashboard

Jan 2026 – May 2026

Python, FastAPI, React, WebSockets, EV Battery Analytics

- Built a real-time EV battery monitoring system showing battery health, charging behavior and operating parameters live to engineers.
- Developed telemetry pipelines for State of Charge (SoC), State of Health (SoH), temperature and charging-cycle analytics, streamed over WebSockets (TCP-based, low-latency push).
- Implemented charging simulations and visualizations to analyze battery behavior across charging scenarios.
- Optimized rendering for high-frequency data updates, linking software interfaces to hardware monitoring.

EDUCATION

B.E. Electronics & Communication Engineering, Jawaharlal Nehru National College of Engineering, Shivamogga (VTU) Aug 2026

CGPA: 7.60 | Class XII (Computer Science): 87% (2021) | Class X: 92% (2019)

CERTIFICATIONS & ACTIVITIES

FutureSkills Prime Certification (NASSCOM) | NITTE Hackathon (2025) | CMRIT Hackathon (2025) | Karnataka State-Level Sports Tournament | Languages: English, Kannada, Hindi